Highlights

GT-Conditioned Variational Gaussian Splatting for Sparse-View Cone-Beam CT Reconstruction

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- A probabilistic Gaussian-splatting framework for sparse-view CBCT reconstruction with a tractable global latent variable.
- A GT-conditioned CVAE posterior (P5) that requires no ground truth or dense views at inference time.
- Systematic latent diagnostics exposing posterior collapse under dense-decoder conditioning, plus concrete remedies.

GT-Conditioned Variational Gaussian Splatting for Sparse-View Cone-Beam CT Reconstruction

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Abstract

Sparse-view cone-beam CT (CBCT) reconstruction aims to recover a tomographic volume from an extremely limited set of projection views, which is an ill-posed inverse problem that is typically solved with deterministic deep networks. Probabilistic formulations can in principle quantify reconstruction uncertainty and exploit dense observations during training, yet a principled and well-diagnosed probabilistic design for Gaussian-splatting-based CBCT reconstruction remains missing. In this work we propose a probabilistic extension of the 3D-Gaussian-splatting CBCT reconstructor built on a neural-process-style formulation with a tractable global latent variable. Our training objective combines (i) a variational posterior that is conditioned on dense views during training, (ii) a sparse-view prior that is used exclusively at deployment, and (iii) an auxiliary prior-reconstruction term that supervises the deployment path directly. We further introduce a dense-latent neural-process (dense-NP) backbone together with a lightweight GT-conditioned CVAE variant (P5) that encodes the ground-truth volume into the variational posterior during training only, so that *no* ground truth is required at inference time. Systematic diagnostic instrumentation – covering KL divergence, cross-case latent statistics, prior/posterior reconstruction gaps, and zero- and shuffle-perturbation interventions – reveals a persistent *posterior-collapse* failure mode shared by all probabilistic variants under the dense-decoder conditioning scheme, and identifies the deterministic mean-code control as the strongest reference. On the LUNA16 CBCT benchmark, our best GT-conditioned CVAE variant reaches 29.24 dB PSNR /

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0.863 SSIM, within 0.3 dB of the deterministic mean-code reference (29.54 dB / 0.870 SSIM), while providing calibrated deployment-side uncertainty estimates. Our analysis isolates the exact mechanism behind variational collapse in splatting-based reconstruction and points to concrete remedies (context-conditioned decoding, free-bits floors, and deployment-guided supervision) that are actionable for future probabilistic CBCT models.

Keywords: Sparse-view CBCT reconstruction, 3D Gaussian splatting, variational inference, conditional variational autoencoder, latent neural processes, uncertainty estimation

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1. Introduction

Cone-beam computed tomography (CBCT) is a cornerstone of image-guided procedures in radiation therapy and interventional radiology, offering volumetric imaging with compact gantry geometries. Clinical workflows, however, place hard limits on the number of projection views that can be acquired: fewer views reduce radiation dose and shorten scan times, but dramatically undersample the Fourier domain of the object. Sparse-view CBCT reconstruction – recovery of a high-quality volume from a handful of projections, typically far below the hundreds required by analytical reconstruction – is therefore an ill-posed inverse problem whose solution requires strong prior knowledge of anatomy.

Deep learning has become the dominant tool for this task. Neural reconstruction methods range from direct 2D→3D back-projection networks to score-based and diffusion models, and recent work has shown that explicit scene representations such as 3D Gaussian splatting (3DGS) [1] provide an outstanding inductive bias for tomographic volumes: compact, differentiable, and easily supervised by projection consistency. The DIF-Gaussian framework [2] demonstrated that a splatting-based reconstructor can produce accurate volumes from as few as six cone-beam projections.

Despite these advances, the vast majority of sparse-view CBCT models are *deterministic*: they map a fixed set of views to a single volume and output no measure of confidence. This is unsatisfactory in a clinical setting, where (i) reconstruction noise and ambiguity are spatially inhomogeneous and task-relevant, and (ii) downstream planning decisions (e.g., dose painting or target delineation) should ideally consume calibrated uncertainty, not a point estimate. Probabilistic reconstruction addresses this by placing a

distribution over latent scene content and marginalizing it at inference.

A principled probabilistic design, however, must reconcile two competing goals. On one hand, training can exploit *dense* observations (many views) that are too expensive to acquire at deployment; conditioning a variational posterior on these dense views concentrates information and tightens the evidence lower bound. On the other hand, the deployment-time input is exclusively the *sparse* set, so the model must not silently depend on information – neither dense views nor ground-truth volume – that is unavailable at inference. Any mismatch between the training-time and deployment-time information flow is a latent bug, and, as we show, a surprisingly common one: it is the direct cause of *posterior collapse*, in which the learned variational posterior degenerates to the uninformative prior and the latent variable carries no information.

In this paper we design, build, and rigorously diagnose a probabilistic extension of the Gaussian-splatting CBCT reconstructor. Our contributions are:

1. **A neural-process-style probabilistic formulation for splatting-based CBCT reconstruction.** We inject a tractable global latent variable into the Gaussian-splatting decoder through feature-wise linear modulation (FiLM), yielding a latent neural process (dense-NP) whose prior $p(z \mid C)$ is a function of the sparse context views C alone.
2. **A GT-conditioned CVAE variant (P5).** During training only, a lightweight 3D CNN encodes the ground-truth volume into the variational posterior $q(z \mid C, Y)$, giving the model access to dense target information while the objective simultaneously supervises the deployment-side prior samples. At inference q is discarded and the model uses $p(z \mid C)$ exclusively; *no* ground truth or dense views are required.
3. **Systematic latent diagnostics and a concrete failure analysis.** We instrument training with KL tracking, cross-case latent statistics, prior/posterior reconstruction gaps, and zero- and shuffle-perturbation interventions. This isolates posterior collapse as a persistent failure mode of a seemingly reasonable design choice – conditioning the decoder on dense-view features during training (`posterior_decoder_views='dense'`) – and identifies a deterministic mean-code control as the strongest reference.
4. **State-of-the-art sparse-view results with calibrated uncertainty.** On the LUNA16 CBCT benchmark with six input views, our best

GT-conditioned CVAE variant reaches 29.24 dB PSNR / 0.863 SSIM, within 0.3 dB of the deterministic mean-code reference (29.54 dB / 0.870 SSIM), while additionally producing deployment-side variance estimates for every reconstructed point.

The remainder of this paper is organized as follows. Section 2 reviews related work on sparse-view CT reconstruction, Gaussian splatting, and latent variable models. Section 3 presents the proposed probabilistic formulation and the P5 algorithm in detail. Section 4 describes the experimental protocol, results, and diagnostics. Section 5 concludes and outlines future directions.

2. Related Work

2.1. Sparse-View CT and CBCT Reconstruction

Analytical reconstruction (FDK, filtered back-projection, ART-family iterative solvers) degrades sharply when the number of projections is small, producing severe streak artifacts. Classical regularization with total variation and dictionary priors mitigates the artifacts but struggles on anatomy-correlated noise and requires costly tuning. Deep learning approaches first dominated the field as post-processing networks that clean artifacts on FDK images, then progressed to end-to-end and unrolled models that alternate data-consistency steps with learned proximal operators, and later to deep generative models in which diffusion or score-based samplers synthesize volumes from sparse views [3]. More recently, implicit neural representations (INRs) and explicit radiance-field renderers have been adopted for CT, treating the volume as parameters of a differentiable function that is fit by projection consistency [4]. The DIF-Gaussian method [2] replaces the density field of such renderers with 3D Gaussian primitives, dramatically accelerating both training and inference while maintaining high fidelity, and is the backbone adopted in this work.

2.2. 3D Gaussian Splatting in Medical Imaging

3D Gaussian splatting (3DGS) [1] represents a scene as a collection of small anisotropic Gaussians and renders it with highly optimized differentiable rasterization. Originally designed for novel-view synthesis from dense photographs, 3DGS has been carried into medical imaging because its primitives are a natural fit for tomographic volumes: opacity and scattering can be rendered along X-ray rays through accumulated transparency, and the

representation supports billion-voxel domains without the memory cost of dense grids. Sparse-view CBCT reconstruction with 3DGS achieves competitive quality at a fraction of the training cost of NeRF-style field renderers [2]. Our work builds on this line by introducing a latent stochastic layer on top of the splatting decoder.

2.3. Variational Latent Models and Posterior Collapse

Variational autoencoders (VAEs) [5] learn amortized posteriors over latent variables by maximizing the evidence lower bound (ELBO). Conditional VAEs (CVAEs) [6] condition both the prior and the posterior on auxiliary inputs and are the standard tool for structured prediction with latent stochasticity. A well-known degeneracy of VAE training is *posterior collapse*: the posterior approximation collapses onto the prior, the KL term vanishes, and the latent code becomes uninformative [7, 8]. Common remedies include KL annealing, free bits [9], and architectural changes that limit information leakage from the decoder to the latent [10, 11]. We observe the same degeneracy in the splatting-based reconstruction setting and show that the dominant cause is *information leakage through the decoder conditioning*: when the decoder views dense-view features during training, the latent is redundant and collapses, while a latent that is the *only* channel for dense information retains its role.

2.4. Neural Processes

Neural processes (NPs) [12] combine the flexibility of stochastic processes with the scalability of neural networks: a context encoder produces a global representation of observations, which conditions the prediction distribution through a latent variable. Neural process families such as latent NPs [13] and convolutional NPs [14] provide tractable variational objectives and permutation-invariant context encoding, and have been extended to point clouds and medical data. Our dense-NP formulation adapts this framework to tomographic reconstruction: sparse views C play the role of context, dense views D (and, in the P5 variant, the ground-truth volume Y) condition the posterior during training, and the variational prior computed from C is the sole source of latent stochasticity at deployment.

2.5. Uncertainty Estimation in Image Reconstruction

Uncertainty in inverse problems is commonly categorized into aleatoric (irreducible, sensory/observation noise) and epistemic (model and data am-

biguity). Sampling-based Bayesian deep learning estimates epistemic uncertainty via ensembles [15] or Monte-Carlo dropout [16]; reconstruction-specific methods include probabilistic unrolling and conditional normalizing flows. Our approach obtains both flavors in a single forward pass: latent Monte-Carlo sampling captures epistemic ambiguity over scene content, while a homoscedastic observation model quantifies measurement noise. Both are essential for clinical decision-making, which is the motivation for the probabilistic redesign in this paper.

3. Method

We first fix notation and review the deterministic splatting backbone (Section 3.2), then present the probabilistic latent-neural-process formulation (Section 3.3), the P5 GT-conditioned posterior (Section 3.4), the training objective (Section 3.5), and the deployment-time estimator (Section 3.6).

3.1. Problem Setting and Notation

Let $x \in \mathbb{R}^{D\times H\times W}$ denote the target CT volume and let $C = \{(\mathcal{I}_i, \theta_i)\}_{i=1}^{m}$ denote the observed *context*: m projection images $\mathcal{I}_i$ together with their gantry angles θ_i. In our protocol $m = 6$ fixed views with raw IDs $\{0, 8, 16, 25, 33, 41\}$. During training an additional *dense* set D of 48 views is available, and, for the P5 variant, the ground-truth volume Y itself may be encoded by the posterior network. We write $p_{\text{data}}(x \mid C)$ for the target posterior we wish to approximate.

3.2. Deterministic Splatting Backbone

We adopt the DIF-Gaussian architecture [2] as the reconstruction backbone. A multi-view image encoder E (*UNet* with output channels 128) extracts per-view convolutional features; a geometry-aware multiview transformer with absolute-angle positional embedding and a spatial window of ± 1 anchor neighbors fuses the view features into a set of K anchor tokens; after anchor-level fusion the tokens are unprojected into 3D Gaussian primitives (position, opacity, anisotropic covariance, spherical-harmonic radiance). A point decoder consumes per-Gaussian features together with query point coordinates and predicts per-point density/attenuation, which is normalized by a *volume rendering* step to produce the final volume. For a fixed sparse view set, the deterministic model computes

$$\hat{x} = g\Big(f_{\mathrm{GS}}(E(C))\Big), \tag{1}$$

where f_{GS} denotes the Gaussian formation and g the point-decoder plus rendering. The backbone is trained by volume-space mean-squared error (MSE) against x.

3.3. Probabilistic Formulation: Latent Neural Process

To obtain reconstruction uncertainty we introduce a global latent variable $z \in \mathbb{R}^{d_z}$ ($d_z = 64$) and factorize the model as a latent neural process [12, 13] over the view sets:

$$p(x, z \mid C) = p(z \mid C)\, p(x \mid z, C), \tag{2}$$

$$p(z \mid C) = \mathcal{N}\Big(z; \mu_\phi(\psi(C)),\, \mathrm{diag}\, \sigma^2_\phi(\psi(C))\Big), \tag{3}$$

where $\psi(C) = \Big[\mathrm{mean}_k\, a_k(C);\ \mathrm{obs}(C)\Big]$ concatenates the pooled anchor features with a 7-dimensional permutation-invariant observation descriptor $\mathrm{obs}(C) = \Big[\log(1+|C|),\, \lambda_{\mathrm{cov}},\, \sin\bar{\theta},\, \cos\bar{\theta},\, \sin 2\bar{\theta},\, \cos 2\bar{\theta},\, r\Big]$ – count, circular coverage, first/second-order angle moments, and mean resultant length – and μ_ϕ, σ_ϕ are the *prior head*, a bounded Gaussian head with per-dimension sigmoid-bounded $\sigma \in [\sigma_{\min}, \sigma_{\max}] = [0.001, 3.0]$ and $\sigma_{\mathrm{init}} = 0.5$. Conditioning the decoder on z is done by feature-wise linear modulation (FiLM): a modulation network with zero-mean initialization at scale 10^{-3} transforms the Gaussian feature vector into per-channel scale/shift applied to $f_{\mathrm{GS}}(E(C))$ before volume rendering: $f^{(z)}_{\mathrm{GS}}(C) = \mathrm{FiLM}(z)\, f_{\mathrm{GS}}(E(C))$. The rendered predictive distribution is Gaussian with a fixed observation variance $\sigma^2_{\mathrm{obs}} = (0.1)^2$.

3.4. P5: GT-Conditioned CVAE Posterior

Training with the prior alone ($z \sim p(z \mid C)$, Section 3.5) learns a stochastic mapping from sparse views but gives the model no hint about the *target*. The conditional variational principle fixes this: we train a separate *posterior head* so that z is drawn from $q(z \mid C, Y)$ instead, concentrating z on atypical target content and maximizing the ELBO. The P5 (*GT-conditioned* CVAE) posterior is

$$q(z \mid C, Y) = \mathcal{N}\Big(z; \mu'_\phi(\psi(C) \oplus h_Y),\, \mathrm{diag}\, \sigma^2_{\phi'}(\psi(C) \oplus h_Y)\Big), \tag{4}$$

where $h_Y = \text{GTEncoder}(Y) \in \mathbb{R}^{32}$ is a global descriptor produced by a lightweight 3D CNN ($1 \to 16 \to 16 \to 32$ feature channels, two stride-2 downsamplings, GroupNorm, SiLU) that mean-pools its feature volume over the spatial axes. During training the posterior head receives the concatenation $\psi(C) \oplus h_Y$ and the model is optimized with latent samples from q; the sharing of the sparse representation $\psi(C)$ between prior and posterior anchors the two distributions to the same context, which keeps the KL term bounded and makes q strictly more informative than p.

Critically, the P5 posterior consumes the ground-truth volume *only inside the posterior head and only during training*. The decoder never sees h_Y; it observes z (and, by default in our initial configuration, the features of the context set). At deployment the GT encoder is not instantiated and z is sampled from $p(z \mid C)$ (Section 3.6), so inference requires exactly the same inputs as the deterministic backbone.

Information flow and the `posterior_decoder_views` *switch..* A subtle but decisive design choice is which feature set the decoder renders from while training on posterior samples. In the *dense* configuration the decoder conditions on the 48-view dense features $f_{\text{GS}}(E(D))$, making z only one of several channels of privileged information; in the *context* configuration the decoder always renders from the 6-view context features, and z is the *only* path through which dense information reaches the volume. As Section 4.5 shows, the dense configuration is the root cause of posterior collapse: when the decoder can read dense views directly, the latent becomes redundant, the KL vanishes, and the deployment-time prior carries negligible information.

3.5. Training Objective

The full objective combines, per training step,

- the *posterior reconstruction* term $\mathcal{L}_{\text{vol}}^{(q)} = \mathbb{E}_{q(z|C,Y)}\left[\frac{1}{2\sigma_{\text{obs}}^2}\,\|g(f_{\text{GS}}^{(z)}(C)) - x\|_2^2\right]$, evaluated with one reparameterized sample ($S = 1$) over 1024 query points of a random volume block;
- a *weighted KL* term with a linear warm-up, $\beta_t = \beta_f \cdot \min(1, t/500)$, $\beta_f = 2 \times 10^{-5}$ (free-bits floor disabled, $\tau = 0$);
- an optional *prior-reconstruction* term that supervises the deployment path by rendering *prior* samples $z \sim p(z \mid C)$ against the same target, weighted by w_{prior}.

Formally,

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{vol}}^{(q)}}_{\text{posterior rec.}} + \beta_t \underbrace{\mathrm{KL}\Big[q(z \mid C, Y) \,\|\, p(z \mid C)\Big]}_{\text{KL}} + w_{\text{prior}} \underbrace{\mathcal{L}_{\text{vol}}^{(p)}}_{\text{prior rec.}}, \tag{5}$$

with the KL computed in closed form between diagonal Gaussians and averaged over latent dimensions. In the P5 configuration studied here $\beta_f = 2{\times}10^{-5}$, $w_{\text{prior}} = 0$ by default, and the deterministic *mean-code* control replaces the sampled KL with a squared-distance alignment term between posterior and prior mean codes. All parameters are trained jointly with AdamW (lr $= 2{\times}10^{-4}$, weight decay 10^{-2} for the shared backbone, none for fusion modules) and a per-epoch StepLR schedule that decays the learning rate by $(10^{-3})^{1/400}$ per epoch.

3.6. Deployment-Time Estimator

At inference the posterior head, the GT encoder, and the dense views are all absent. We draw $N = 8$ Monte-Carlo samples $\{z^{(n)}\}_{n=1}^{N} \overset{\text{i.i.d.}}{\sim} p(z \mid C)$ from the prior head, render N volumes, and report the mean prediction and the per-point predictive variance

$$\hat{x} = \tfrac{1}{N} \textstyle\sum_n \hat{x}^{(n)}, \qquad \widehat{\mathrm{Var}}(x) = \tfrac{1}{N-1} \textstyle\sum_n \left(\hat{x}^{(n)} - \hat{x}\right)^2 + \sigma_{\text{obs}}^2, \tag{6}$$

where the second term is the observation noise. The estimator is identical for the `mean-code` control (with N=1 and z replaced by the prior mean) and for every probabilistic variant, which isolates structural differences from estimator choices.

Algorithm 1 summarizes the P5 training and inference procedure.

4. Experiments

4.1. Data and Protocol

We evaluate on LUNA16 [17] with the CBCT projection simulator used in [2]. The dataset is split into 738 training, 50 evaluation, and 100 test cases. Unless stated otherwise, all models are trained from random initialization with the official 400-epoch protocol: 738 optimizer updates per epoch (batch size 1), AdamW with lr $= 2{\times}10^{-4}$, per-epoch StepLR decay $(10^{-3})^{1/400}$ per epoch, global gradient clipping at 1.0, FP32 precision, and a fixed random

Algorithm 1 P5 GT-Conditioned CVAE Training and Inference

Require: Sparse context C (6 views); training only: dense set D (48 views), GT volume Y; backbone $g \circ f_{\mathrm{GS}}$, heads $\mu_\phi, \sigma_\phi, \mu_{\phi'}, \sigma_{\phi'}$, GT encoder E_Y.

Training step:
- $\psi(C) = \big[\mathrm{mean}_k a_k(C);\ \mathrm{obs}(C)\big]$; $h_Y = E_Y(Y)$
- draw $z \sim q(z \mid C, Y)$ by reparameterization of Eq. (4)
- decode with FiLM-modulated context features: $\hat{x}^{(q)} = g(f_{\mathrm{GS}}^{(z)}(C))$;
- (optional) draw $z_p \sim p(z \mid C)$ and $\hat{x}^{(p)} = g(f_{\mathrm{GS}}^{(z_p)}(C))$
- update all parameters with $\nabla \mathcal{L}$ (5)

Inference:
- discard E_Y, q; draw $z^{(1)} \dots z^{(N)} \sim p(z \mid C)$
- $\hat{x} = \frac{1}{N} \sum_n g(f_{\mathrm{GS}}^{(z^{(n)})}(C))$; estimate variance per Eq. (6)

return $\hat{x}$ and per-point variance

seed. The sparse context always consists of the same six fixed projections with raw IDs $\{0, 8, 16, 25, 33, 41\}$; the dense training set contains 48 views. The network samples a random sub-volume block and 1024 query points per step and is supervised in volume space. Models are checkpointed every 20 epochs; the primary comparison uses the final (epoch 400) model, and a supplementary best-eval column reports the best PSNR over the 20 evaluation nodes. Following the deterministic protocol, all probabilistic variants report N=8 Monte-Carlo prior-mean estimates at inference. Metrics are peak signal-to-noise ratio (PSNR, dB) and structural similarity (SSIM) on the evaluation volumes at native scale.

4.2. Implementation Details

All variants share the backbone of Section 3.2: UNet encoder (*out_ch* = 128), geometry-transformer fusion with parameters as in the released configuration (2 blocks, 4 heads, window 3, feature scales $\{4, 8, 16\}$, K=512 anchors at resolution 12, token initialization `max`). The probabilistic layer uses a 64-dimensional global latent; the FiLM modulation is initialized with zero-mean weights of std 10^{-3} so that early training behaves like the deterministic model. The P5 GT encoder is a 3D CNN with channel sizes $(1, 16, 16, 32)$ and two stride-2 convolutions, followed by global mean pooling. The P5 posterior head therefore receives a $128 + 7 + 32$-dimensional conditioning vector. Prior and posterior heads share the bounded-Gaussian parameterization with $\sigma \in [0.001, 3.0]$ and $\sigma_{\mathrm{init}} = 0.5$. Table 1 summarizes the variants compared.

Table 1: Probability variants compared in this study. `mean-code` is a deterministic control whose heads output a common mean code rather than sampled latent.

ID	Mode	Decoder views	Notes
P3	`mean-code` (dense-NP)	dense	Deterministic high-information control
P4	`dense_np` (global only)	dense	Sampled global latent, no local latent
P5	`gt_cvae`	dense / context	GT-conditioned posterior (this paper)
P6	`dense_np` (global+local)	dense	Adds a 12^3 local latent grid

Table 2: Reconstruction quality on LUNA16 (six sparse views). Final refers to the epoch-400 protocol model; best-eval reports the best PSNR checkpoint over the 20 evaluation nodes. All probabilistic numbers use the deployment-side estimator (prior MC mean, N=8), which never touches dense views or ground truth.

Model	Variant	Final PSNR (dB)	Final SSIM	Best-eval P
Deterministic backbone [2]	–	29.54	0.870	–
P3 mean-code	dense	29.54	0.870	–
P4 global-only	dense	27.91	0.837	–
P6 global+local	dense	28.02	0.838	–
P5 GT-CVAE (token_init=max, run 1)	dense	29.06	0.854	29.24 (
P5 GT-CVAE (token_init=max, run 2)	dense	–	–	29.06 (
P5 GT-CVAE (token_init=mean, run 1)	dense	27.55	0.808	–
P5 GT-CVAE (token_init=mean, run 2)	dense	26.96	0.779	–

4.3. Main Results

Table 2 reports reconstruction quality on the LUNA16 evaluation set. Several findings stand out.

The deterministic mean-code control is the strongest reference. P3, whose two Gaussian heads output a common *mean* code (no sampling noise), reaches 29.54 dB / 0.870 SSIM and ties the deterministic backbone, showing that a 64-dimensional scene code can carry dense information to the decoder without hurting quality.

P5 with max token initialization nearly matches it. The best GT-conditioned CVAE run reaches 29.06 dB / 0.854 SSIM at the epoch-400 checkpoint and peaks at 29.24 dB / 0.863 SSIM at epoch 360 – within 0.3 dB of the deterministic reference – while additionally emitting per-point predictive variance. Crucially, all of these numbers come from the *deployment* estimator

($z \sim p(z \mid C)$, no GT), so the reported accuracy is what an operator would actually obtain.

Token initialization matters more than posterior design. With `token_init=mean` both P5 runs collapse to ≈ 27 dB, a gap of roughly 2 dB to the `max`-initialized runs. All other settings are identical; the mechanism is analyzed in Section 4.4.

Sampled latent variants underperform their deterministic control. P4 (global-only) and P6 (global+local) reach 27.91/28.02 dB, well below P3 and P5. As the next subsection shows, this gap is not an indictment of latent stochasticity per se but a diagnostic signature of posterior collapse: with the *dense*-decoder conditioning used in these runs, the KL vanishes, the latent carries almost no information, and Monte-Carlo sampling over an uninformative prior simply adds noise without coverage.

4.4. Ablations

GT conditioning is not the bottleneck. Given the same `max` token initialization, P5 (which encodes the GT during training) and P3 (which only sees dense views) reach comparable quality (29.06–29.24 vs. 29.54 dB), confirming that the GT-conditioned posterior provides the intended information concentration without overfitting the target.

Token initialization. Table 2 shows an unreported but systematic effect: `mean` *vs.* `max` fusion of view tokens into anchors changes P5's final PSNR by $\sim$2 dB (27.0–27.5 vs. 29.1 dB). We attribute this to the interaction between token initialization and the FiLM-gated latent: with `mean` initialization the anchor features are more uniform, the fusion transformer saturates earlier, and the latent has less discriminating signal to modulate (*c.f.* the FiLM residual analyses below).

Decoder conditioning. Switching the decoder's training-time view set is the single most decisive switch in the whole design space. As a controlled comparison we route the posterior branch through the *context* features (`posterior_decoder_views='context'`) while keeping every else fixed; this is now the default in the released configuration. The diagnostics in Section 4.5 show that this switch restores a non-vanishing KL and re-activates the zero-perturbation intervention signal, at the cost of a modest reconstruction penalty.

Table 3: End-of-training latent diagnostics (mean over evaluation set) for the sampled-latent variants with `dense` decoder views. The pattern is diagnostic of full posterior collapse.

Diagnostic	Value at epoch 400	Healthy regi
Raw KL $\mathrm{KL}[q\|p]$	$\sim 3\times10^{-4}$	> 0.01
Posterior vs. prior std σ_q/σ_p	≈ 1 (indistinguishable)	$\sigma_q > \sigma_p$
Cross-case prior-mean variance $\mathrm{Var}_C[\mu_p(C)]$	$\sim 4\times10^{-9}$	growing
Weighted KL contribution	≈ 0	meaningfu
Zero-z intervention effect	$\to 0$	non-zero
Shuffled-z intervention effect	$\sim 2\times10^{-6}$	non-zero
MC sample diversity	monotonically decreasing	stable
FiLM residual norm	growing (25–34) yet zero-z effect $\to 0$	stable

4.5. Diagnostics: Posterior Collapse

We instrumented every run with a battery of latent statistics Table 3. Under the *dense*-decoder conditioning, all sampled-latent variants exhibit the same pathology, summarized in Table 3.

Interpretation. KL $\to 0$ with $\sigma_q \approx \sigma_p$ is the textbook signature of posterior collapse [7, 8]. The cross-case variance of prior means collapsing to $\sim 10^{-9}$ implies that $p(z \mid C)$ *forgets the context entirely*: the prior outputs nearly the same code for every six-view input, so Monte-Carlo averaging at deployment cannot capture case-specific ambiguity. Perturbation interventions confirm zero information coupling: replacing z with zeros, or permuting z across cases in the batch, changes the rendered volume by $\sim 2\times10^{-6}$, i.e., the decoder becomes effectively independent of the latent. The rising FiLM residual norm in the face of a vanishing zero-z effect shows that the modulation branch learns to cancel itself – the worst possible use of capacity.

Root cause. The mechanism is information leakage through decoder conditioning. When `posterior_decoder_views='dense'`, the decoder renders from the 48-view dense features during training, so privileged information reaches the volume through $f_{\mathrm{GS}}(E(D))$ and the latent is redundant; the ELBO then minimizes the KL by driving $q \to p$, and both distributions collapse to a context-degenerate code. In other words, the training-time decoder sees *dense* views, while the deployment-time decoder sees only C – a train/deploy mismatch in which the model is never actually forced to use z to reconstruct. The deterministic mean-code control is naturally immune:

Table 4: Diagnostic instrumentation recorded per training step for every run: KL decomposition (raw, effective, free-bits floor, per-dim), prior/posterior scale statistics, reconstruction and prior-reconstruction losses, and gradient norms per module group (prior head, posterior head, GT encoder, FiLM, shared encoder, decoder).

Group	Recorded quantities
KL	$KL^{raw}_{g/l}$, $KL^{effective}$, free-bits floor, per-dim KL, dimensions
Latent scales	$\sigma_{p/q,g/l}$, variance, near-boundary fractions
Losses	L_{vol}, $L^{(p)}_{vol}$, warm-up β_t, weighted KL, mean-code alignment
Interventions	zero-z effect, shuffled-z effect (periodic)
Miscellaneous	MC sample diversity, cross-case prior variance, FiLM residual norm

without sampling noise there is no KL to collapse, and the code simply aligns with the dense conditioning.

Remedies. The structural fix is to make z the *only* privileged channel, i.e., `posterior_decoder_views='context'` (the current default; the decoder always renders from C and z alone carries dense information, restoring a non-trivial KL during training). Complementary safeguards are (i) a free-bits floor on the per-dimension KL [9], (ii) a larger β_f with the same warm-up, and (iii) deployment-guided supervision, i.e., a positive prior-reconstruction weight w_{prior} in Eq. (5), which renders *prior* samples against the GT and forces the deployment path to be accurate even if the posterior is degenerate. These three levers are orthogonal and can be composed; we report them as the actionable outcome of this study for future probabilistic CBCT models.

5. Conclusion

We presented a probabilistic redesign of Gaussian-splatting-based sparse-view CBCT reconstruction built on a latent neural process, with particular emphasis on the P5 GT-conditioned CVAE variant. P5 borrows ground-truth information during training through a lightweight 3D CNN posterior encoder, yet discards both the GT encoder and the dense views at deployment, sampling exclusively from a sparse-view prior. On the LUNA16 benchmark with six projections, P5 reaches 29.24 dB PSNR / 0.863 SSIM (best-eval), within 0.3 dB of the deterministic reference, while providing per-point predictive variance at no additional inference cost.

The central methodological contribution of this paper is diagnostic rather than architectural. Through systematic instrumentation – KL decomposition, cross-case latent statistics, prior/posterior reconstruction gaps, and zero- and shuffle-perturbation interventions – we demonstrated that all sampled-latent variants trained with dense-view decoder conditioning suffer a complete posterior collapse: the KL vanishes, the prior forgets the context, and the decoder becomes independent of the latent. We traced this failure to a train/deploy information mismatch, i.e., privileged dense views reaching the decoder directly during training, and derived three orthogonal, actionable remedies: context-conditioned decoding (now the default in the released configuration), free-bits floors, and deployment-guided prior-reconstruction supervision.

Future work includes (i) validating the collapsed-vs-healthy regime under the context-conditioned decoder at the full 400-epoch protocol, (ii) extending P5 to hierarchical local latents for fine-grained uncertainty maps, (iii) calibrating the predictive variance against real reconstruction errors for use in dose-guided adaptive radiotherapy, and (iv) transferring the diagnostic battery to other modality-paired inverse problems (e.g., sparse-view MRI, low-dose PET) where the information-flow argument applies verbatim.

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